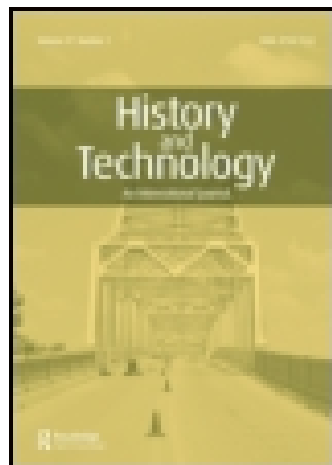




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SILICON FOR INDUSTRY: COMPONENT DESIGN, MASS PRODUCTION, AND THE MOVE TO COMMERCIAL MARKETS AT FAIRCHILD SEMICONDUCTOR, 1960–1967

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Abstract: Most accounts of the microelectronics revolution have emphasized the role of military patronage and procurement in the shaping of silicon technology and the consolidation of the semiconductor industry. Little attention has been devoted, however, to the silicon industry's shift from military to commercial markets in the early and mid-1960s. Drawing on an examination of Fairchild Semiconductor, the firm that initiated this shift, this essay argues that the silicon industry's expansion into non-military markets was indissociable from deep changes in manufacturing, organizational structures as well as component and system technologies. Special attention is devoted to the ways in which Fairchild created a user base for its products in the computer and consumer electronics industries by hiring engineers from these sectors and encouraging them to design components as well as applications for the firm's potential customers. This article also examines how Fairchild introduced mass production techniques from the automotive and electrical industries, delocalized its assembly operations to South-East Asia, and developed new packaging techniques to meet the cost and manufacturing requirements of commercial users. This article is intended to contribute to recent interest in the role of user – supplier relations in technological change and, more specifically, to the question of the pervasive use of silicon components across a wide range of industrial sectors, a central phenomenon in the history of industrialized economies since the 1960s.

INTRODUCTION

The silicon industry underwent major changes from the mid-1950s to the 1970s. It grew tremendously, with sales exploding from \$17

million in 1957 to \$8.6 billion in 1978. The silicon industry became also central to the political economy of American manufacturing. Whereas silicon firms served almost exclusively the military and aerospace market in the late 1950s, they had become, twenty years later, key suppliers to most industrial sectors in the US economy such as computer, automotive, telecommunication, and consumer electronics. More importantly, the widespread adoption of silicon transistors and integrated circuits led to major changes in system design across a wide range of industries and brought about generalized productivity gains throughout the economy. How did the silicon industry gain such a central place in the US economy? How did silicon microelectronics become what economist Manuel Trajtenberg calls a "general purpose technology," a technology "characterized by [its] pervasive use in a wide range of industrial sectors and by [its] technological dynamism"?¹

To explore these questions, this essay investigates the ways in which Fairchild Semiconductor transformed microelectronics into a general purpose technology. Now almost forgotten, Fairchild Semiconductor, a firm located on the San Francisco Peninsula, played a key role in the history of the silicon industry. It developed the main design and process technologies in silicon electronics such as the planar process, the integrated circuit, and the MOS transistor. It was also at the origin of the silicon industry in the "Silicon Valley." Besides training scores of engineers, technicians, and operators in silicon processing, Fairchild spawned more than thirty semiconductor corporations, including Intel, AMD, and National Semiconductor, on the San Francisco Peninsula between 1959 and 1972. More importantly, Fairchild Semiconductor pioneered the construction of mass markets for silicon transistors and integrated circuits in the 1960s. In particular, Fairchild, which at first was oriented nearly exclusively toward military avionics, was the first silicon firm to develop commercial markets for its products. Within a few years, Fairchild became a major supplier of silicon components to commercial users, gaining 80% of the computer market for integrated circuits by 1968. Thereby, Fairchild set a precedent and a pattern which other firms such as Intel, National Semiconductor and Motorola closely followed in the late 1960s and 1970s. Focusing on Fairchild's move to the commercial markets, this paper investigates the ways in which the firm built new markets for its products in the commercial sector. It also examines how Fairchild's market strategies shaped its design technologies and manufacturing systems.

The silicon industry's entry into the commercial markets has received scant attention in the historical and economic literatures. While the military shaping of semiconductor technology has been the center of a lively debate since the mid-1960s, the silicon industry's expansion into the computer and consumer markets has received only passing attention by economists such as William Golding, Richard Levin, and Edward Steinmueller. These authors have emphasized the critical role of learning economies, yield improvements, and the resultant price reductions in the diffusion of silicon technology to non-military sectors. They have not however examined the ways in which the silicon industry acquired its manufacturing expertise. Nor did they explore the social and economic forces that led silicon firms to enter commercial markets. Finally, they did not analyze the ways in which silicon manufacturers constructed novel products and technologies to meet the needs of new users.²

Following recent work on user-supplier relations, this essay argues that Fairchild expanded into the commercial sector in the early and mid-1960s because of changes in military procurement policies and the emerging demand for silicon devices among computer and consumer electronics manufacturers. This article also claims that Fairchild's successful entry into these markets was due to its acquisition of new technological and marketing competences and to its ability to fulfill the needs of computer and consumer electronics users. In particular, Fairchild hired engineers from sectors that were intended as users of its products. It encouraged them to design both components and their applications for potential clients. To meet the imperatives of low cost, high volume, and automatic production exerted by commercial customers, Fairchild also introduced mass production techniques from the electrical and automotive industries and shifted its assembly operations to the Far East. These imperatives, this paper argues, also led Fairchild to develop new types of packages, the protective containers in which silicon chips are enclosed, so as to accommodate the manufacturing needs of the consumer electronics and computer industries.

GOING COMMERCIAL

Fairchild Semiconductor was established in Palo Alto in 1957 by a group of young physicists, chemists, and engineers (Robert Noyce,

Gordon Moore, Jean Hoerni, Jay Last, Sheldon Roberts, Julius Blank, Eugene Kleiner, and Victor Grinich) who had been trained in the new field of silicon electronics by William Shockley, at Shockley Semiconductor Laboratory. Shockley, the co-inventor of the transistor, had set up this new laboratory specializing in the development of silicon transistors on the San Francisco Peninsula in 1955. Rebelling against Shockley's autocratic management style and intent upon controlling their own technical work, these men obtained financial backing from Fairchild Camera and Instruments, a medium-sized military contractor on the East Coast, to establish a firm specialized in the development and manufacture of silicon components for military applications.³

Within a few years, Fairchild Semiconductor radically reshaped the silicon industry by revolutionizing its products and manufacturing methods and by mastering the volume production of very complex silicon components. Fairchild adopted new laboratory techniques such as diffusion and photolithography developed at the Bell Telephone Laboratories to make high precision silicon components and transformed them into reproducible manufacturing processes. As a result, Fairchild was the first commercial firm to introduce high frequency silicon transistors to the market. These transistors met a growing demand for fast silicon components in the emerging field of digital-based guidance and control systems for jet aircrafts and missiles. However, at first Fairchild's transistors did not fully meet the reliability criteria of the Air Force and the Minuteman program. Adapting their technologies to meet those reliability standards, Fairchild's founders, made a second round of major process and design innovations. In particular, Hoerni developed the planar process in 1959 – a key manufacturing method, the main characteristics of which was the growing of a layer of silicon oxide on top of the wafer which protected it from outside contamination and made possible the production of ultrareliable diodes and transistors. Capitalizing on the planar process, Noyce and Last designed a new silicon device, the integrated circuit, by interconnecting individual components in the same wafer through metal wires evaporated on top of the oxide layer. Improving the reliability of electronics systems and permitting their rapid miniaturization, the integrated circuit became the silicon industry's main product in the mid-1960s.⁴

In conjunction with the development of new products and fabrication processes, Fairchild pioneered the quantity production of diffused silicon components. To manufacture these very complex

products in volume, Fairchild, with support from the Minuteman program, developed new manufacturing systems which had little equivalent in other branches of the electronics industry. These systems had four characteristics. First, they were based on a hybrid process with a batch component and a continuous flow element. In the first part of the manufacturing process, transistors were batch produced, as groups of wafers or slices of silicon crystal containing hundreds of transistors were diffused and oxidized at the same time. The second part of the process was an assembly affair, reminiscent of the production lines in the mechanical industries. The wafers were sliced and the individual transistor dice or chips were hand soldered to the base of the transistor packages. Finally, gold wires were attached to the transistor contacts and each unit was hermetically sealed. Second, Fairchild's manufacturing systems were very "process intensive." In other words, the production of diffused silicon components depended on very complex chemical processes with thousands of steps. As the lateral and vertical dimensions of diode and transistor structures counted in the microns, a single error at any process step led to disastrous yields (the percentage of fully operational units emerging from the production process). Third, to achieve such levels of precision and reproducibility, these systems rested on an extraordinarily tight industrial discipline – with no equivalent nor precedent in the history of American manufacturing. Finally, they were characterized by very clean and laboratory-like environments. Because silicon transistors and integrated circuits were very sensitive to contamination, Fairchild's production lines had more stringent cleanliness requirements than the manufacturing areas of other electronics firms.⁵

Because of its revolutionary innovations and its ability to produce advanced components in volume, Fairchild was well positioned to take advantage of the explosion of the military demand for high performance silicon components brought about by Sputnik in the late 1950s. The firm gained a large share of the military market for high reliability silicon components. It saw its sales swell from \$500,000 in 1958 to \$21 million in 1960 and its workforce expand from 12 employees in 1957 to 1,400 three years later.⁶ As a result, Fairchild emerged as the leading silicon manufacture in the United States in the late 1950s and very early 1960s.

Born in the military sector, Fairchild Semiconductor gradually expanded into the commercial markets in the early and mid-1960s. No existing records can help reconstruct Fairchild's decision to enter

the commercial markets. However, there is strong evidence that Robert Noyce, Richard Hodgson, Gordon Moore, and others at Fairchild Semiconductor reoriented the firm towards non-military markets because of increasing competition in the silicon industry. They also became increasingly aware of profound changes in military procurement, and, more importantly, they saw the rising demand among computer and consumer electronics manufacturers for very high performance silicon devices.

Because of increasing competition from other silicon corporations, Fairchild, which had a virtual monopoly on the production of advanced silicon transistors in the late 1950s, saw its dominant market position in the military sector challenged in succeeding years. While large East Coast firms such as Philco, RCA, and General Electric were slow to adopt Fairchild's innovations, other firms, benefiting from the courts' loose enforcement of intellectual property rights, entered the manufacture of planar transistors and integrated circuits in the early 1960s. Motorola and Texas Instruments (TI), for instance, set up crash research and engineering programs in planar technology in 1959. They established large production facilities for the manufacture of advanced silicon devices in the following years. Fairchild Semiconductor faced also competition from its own "spin-offs" such as Amelco, Signetics, and General Microelectronics. Interested in further developing transistor and integrated circuit technology, five groups of Fairchild engineers and managers left the company to establish their own silicon corporations in the late 1950s and early 1960s. These firms exploited Fairchild Semiconductor's design and process innovations, and rapidly gained a substantial share of the market for advanced silicon components, especially integrated circuits, in the early 1960s.⁷

While Fairchild faced increasing competition from new as well as established silicon manufacturers, it also confronted deep changes in military procurement. These changes affected its profitability and underscored the perils of relying almost exclusively on the military market. In particular, Secretary of Defense Robert McNamara instituted a severe cost-reduction program and restructured military procurement in the early 1960s. To reduce costs, the Department of Defense (DoD) phased out a number of weapon systems, cancelling some which were already under development and halting the deployment of others. The DoD also reduced its purchasing of parts and subsystems, and eliminated excess inventories. As a result of these cutbacks, military procurement of electronics components

dropped substantially. While silicon manufacturers suffered less from military retrenchment than their counterparts in the germanium and microwave tube industries, they saw their sales brutally halted: The military market for silicon transistors, which had grown explosively from \$1.8 million in 1955 to \$99 million in 1960, declined to \$96 million in 1961 and increased at a slow pace thereafter.⁸

In addition to cutbacks in spending, the Department of Defense reformed the procurement process to effect price reductions. While the DoD had predominantly used sole source contracts in the 1950s, it sought, under McNamara, to foster competition among military contractors. The Department of Defense opened military procurements to competitive bidding. It also sought to create new suppliers by claiming rights to its contractors' intellectual property and manufacturing drawings and by handing them over to competing firms. The DoD also pressed more forcefully for second sourcing in its procurement contracts.⁹ In addition to fostering competition, the Department of Defense also sought to diminish the bargaining power of its suppliers in price negotiations in order to reduce the price of military components and systems. To do so, the DoD increasingly required prime contractors to disclose their manufacturing costs. It also demanded that, in turn, they obtain cost information from their subcontractors and negotiate prices on the basis of these costs. Finally, the DoD increasingly required access to its suppliers' accounting books and audited their cost estimates as well as their actual costs as a way of reducing their profits.¹⁰

McNamara's procurement policies as well as growing competition from TI, Motorola, Amelco, and Signetics had severe repercussions on Fairchild. Defense retrenchment and growing competitive pressures slowed the firm's rapid growth. While Fairchild's sales had exploded from \$500,000 in 1958 to \$21 million in 1960, they grew at a slower pace in the early 1960s. Fairchild also ran into severe inventory overruns. For example, the firm laid off 25% of its workforce in the Spring of 1960 because of lower than expected sales. Finally, the firm, which had previously "supreme bargaining power" in its dealings with customers, discovered to its dismay that it could not influence pricing anymore.¹¹

The DoD's attempts to weaken the bargaining position and reduce the profits of its suppliers also adversely affected Fairchild. Taking advantage of the emergence of new suppliers of advanced silicon transistors and integrated circuits, the DoD and prime military contractors put growing pressure on Fairchild to submit pricing

data and grant access to its books. Increasingly, they asked the firm to accept second sourcing and price redetermination as well as to relinquish rights to its processes and production drawings. While these demands were difficult for most military suppliers to accept, they were particularly unacceptable for Fairchild Semiconductor and its parent company Fairchild Camera and Instruments. Since its founding, Fairchild Semiconductor had tenaciously kept its independence from military oversight. In particular, it had consistently refused military research contracts because these gave the military rights to its technology as well as control on its research and engineering programs. Similarly, Fairchild Camera had traditionally refused recosting and price redetermination on sales of off-the-shelf products. Although Fairchild Semiconductor strongly resisted military demands, the firm increasingly had to "cave in" and, as a result, lost some of its autonomy. "The competition became so great," recalled Nelson Stone, Fairchild Camera's general counsel, "that we had to back off from our policy. We no longer had as many unique products. And more and more companies were agreeing to this sort of thing."¹²

As the military market became very competitive and had more constraints attached to it, Fairchild Semiconductor sought to expand into the commercial markets. Performing commercially-oriented market research, Fairchild's sales and marketing department identified new business opportunities for the firm's high performance devices in the computer and consumer electronics industries. In particular, the marketing division found out that recent regulations and policies of the Federal Communication Commission were opening up a new market for high performance silicon components in television broadcasting. Reasoning that very high frequency (VHF) bands could not accommodate the increasing number of television broadcasting stations, the FCC decided in 1960 to fully exploit the ultra high frequency (UHF) portion of the electromagnetic spectrum. As a result, the FCC required all television monitors to have the capability of receiving UHF channels. This ordinance had the unintended consequence of creating a market for silicon transistors. Because TV tuners using conventional radio tubes could not handle the higher frequencies, television manufacturers got increasingly interested in planar silicon transistors: unlike standard vacuum tubes, planar transistors could easily operate in the UHF range. They also had the added benefit, unlike germanium transistors and RCA's microtubes, of being dependable and temperature resistant,

a characteristic of key importance to the reliability-oriented television industry.¹³

Similarly, competitive pressures in the commercial computer industry led a number of computer manufacturers to switch to planar silicon transistors and diodes in the early 1960s. Because computer architectures had largely stabilized by that time, manufacturers of commercial computers competed increasingly by improving the speed and reliability of logic circuits. As the performance of these circuits depended to a large degree on electronics components, computer manufacturers who had previously used germanium transistors got increasingly interested in planar devices. Planar transistors could withstand higher temperatures than germanium transistors. They also had equal or greater frequency characteristics.¹⁴

Control Data (CDC), a producer of high end computers for computationally demanding applications, is a case in point. In 1961, Control Data switched to planar transistors because of the difficulties it encountered in designing the 6600, a "supercomputer" meant to compete directly with IBM's machines in the scientific computer market. Control Data had designed its computer around standard packaging techniques but ran into very substantial "packaging" difficulties. Early design experiments revealed noise and oscillation problems in the back panel wiring which were deemed unacceptable for a system of that size. The source of these difficulties was that the switching speeds required were so high that wave lengths present in the signals were shorter than the wire lengths in the back panels. In other words the physical size of the computer had become a limitation on speed.¹⁵

To shorten up the critical wiring paths, Control Data made drastic changes in packaging configurations. The firm replaced traditional building blocks with a new package, the cordwood package, which had much higher function densities. The new package used two small printed circuit boards back to back with all components mounted internally. Up to sixty-four transistors could be assembled in each module, a density improvement by a factor of ten over previous packaging techniques at CDC.¹⁶ While these packages solved the computer's noise and oscillation problems, they imposed severe temperature constraints on transistors. Control Data "had built up these modules out of Motorola [germanium] devices," Tom Bay, Fairchild's marketing director, later recalled, "and they were fine on the bench but when you started packing them together the way they had to in the machine, they just generated too much heat and they would

not stay together."¹⁷ As a result, the engineering team designing the Control Data 6600 got interested in planar silicon devices which were more temperature resistant than germanium transistors.¹⁸

The computer and consumer electronics markets not only offered an alternative to military procurement, but they were potentially larger, more stable, and profitable than their military counterparts. The consumer electronics industry, for example, offered a very sizable and fast expanding market as the total production of television sets grew from 5 to 7 million between 1958 and 1963. The commercial computer market was even more promising. Because of growing requirements for computers in science, industry, and government, the commercial computer industry exploded in the early 1960s. Computer shipments swelled from 1,000 in 1957 to 4,500 in 1962. By 1967, the total production of computers had reached 18,700. Because computers such as the Control Data 6600 used more than half a million transistors, the commercial computer industry represented a very large market indeed, one which potentially dwarfed the military sector.¹⁹

ENGINEERING COMPONENTS, ENGINEERING SYSTEMS

To exploit growing opportunities in the commercial sector, Fairchild followed a very innovative approach which other firms, including its own spinoffs, imitated to various degrees in the following years. To expand into the commercial markets, Fairchild Semiconductor needed to acquire a deep knowledge of system engineering. It also needed to expand its marketing expertise into the commercial markets. It began this endeavor by restructuring its sales and marketing organization and by hiring system engineers from sectors intended to be users of its products. While Fairchild Semiconductor had, like other silicon corporations, a relatively small and military-oriented marketing and sales force by 1960, the firm's management, especially Bay, greatly strengthened it in the early and mid-1960s, transforming it into the largest and most effective sales and marketing organization in the silicon industry. Fairchild's management not only very substantially enlarged the sales department but also gave it a strong technical backing by transferring the application engineering laboratory from R&D to marketing in 1961. Application engineering had previously concentrated on characterizing and evaluating new devices for military applications. With this new restructuring, it

increasingly gave technical support to sales and marketing and acted as a technical interface with commercial customers.²⁰

Fairchild also reconfigured its sales and marketing department in the early and mid-1960s. To better understand the needs of commercial users, Fairchild Semiconductor restructured the product marketing department as well as the regional sales offices along market lines in 1963. "When [the company] really started getting into the commercial markets," recalled Thomas Bay, the firm's sales and marketing manager,

we (the management group) reorganized to not only regions but market area. For instance, on the West Coast, we had a computer guy who handled Scientific Data Systems and GE Phoenix and we had a consumer guy. And then we had product marketing organized similarly: we had a product marketing group for each of those markets to keep our finger on what the [users] were looking for and so what products we needed to continue to grow in various market segments.²¹

Two years later, Fairchild Semiconductor, in a very innovative move, restructured its entire sales and marketing department around four markets: computer, consumer, industrial, and military. Enjoying substantial autonomy, each market division had its own sales, product marketing, application engineering, and product planning groups as well as its own legal office to draft production contracts with customers.²²

In conjunction with building a strong commercial marketing and sales department, Fairchild Semiconductor brought the "customers" into the firm to acquire a better understanding of their requirements. In 1962, the firm's management established a high speed memory engineering department in the research laboratory to develop a "memory system". They staffed it with engineers from General Electric's computer division. One year later, they established a digital systems department in the research laboratory under Rex Rice, a computer engineer from IBM, to learn about new computer architectures and packaging techniques. The firm also hired a large number of circuit and system engineers with experience in the computer and consumer electronics industries. They were assigned to "beef up" the device development, marketing, and application engineering departments. Indeed, most engineers in the application engineering laboratory had worked previously at computer and consumer electronics firms before joining Fairchild. In particular,

a large contingent of application engineers had been schooled in computer technology at IBM and Philco's Western Development Laboratory. Others came from Zenith and various consumer electronics companies. As Fairchild Semiconductor brought in system engineers from user sectors and built a strong sales and marketing department, the firm gained a strong system and marketing expertise in the computer and consumer electronics industries.²³

Exploiting its system competence as well as its strong interface with commercial users, Fairchild Semiconductor put substantial efforts into designing high performance components that met the complex circuit and system requirements of computer and consumer electronics manufacturers. To do so, Fairchild closely involved its inhouse application engineers in the definition and design of new products. These application engineers identified the needs of computer and consumer electronics manufacturers by relying on their circuit and system expertise and by conferring with the customers. Based on these requirements, they determined the devices' characteristics. "We sought to understand the circuit and system requirements of the customer," reminisced John Hulme, the head of the application laboratory, "and from there define the product. We would start by asking: What does the system need to do? How can we do it with what we have?"²⁴ Having identified the circuit and system needs of the users, application engineers, in collaboration with the research and development laboratory, designed transistors and integrated circuits that fit the firm's manufacturing processes and met the system needs of commercial users.²⁵

Following this approach to component design, Fairchild developed a broad line of system-oriented transistors, diodes, and integrated circuits in the early and mid-1960s. While the firm had eight military-type transistors on the market in 1960, it manufactured four years later 130 different transistors, most of which were for commercial applications. Similarly, the firm marketed more than one hundred commercially-oriented integrated circuits by the mid-1960s. Each of these components was meant to fit a particular circuit or system niche. For example, Fairchild Semiconductor designed, among others, transistors for saturated switching circuits, non-saturating switching circuits, and amplifier and oscillator circuits. The firm also produced a wide variety of standard integrated circuits to serve as building blocks for systems ranging from stereos to minicomputers.²⁶

Going one step further in its efforts to meet the circuit and system needs of commercial users, Fairchild customized its transistors and,

later, manufactured custom integrated circuits for computer manufacturers. Building on its line of high performance logic transistors, Fairchild's engineers designed very high speed transistors for Control Data's supercomputers. In the mid-1960s, Fairchild's engineers also developed whole families of integrated circuits with flip-flops, gates, and buffers for IBM, Honeywell, and Scientific Data Systems. These corporations would approach Fairchild with proprietary circuits which the firm's engineers would transform to meet the constraints of the silicon process. As more and more computer corporations wanted their own circuits, Fairchild became a major manufacturer of custom products for selected companies.²⁷

In conjunction with the development of system-oriented devices, Fairchild engineered new applications around its diodes, transistors, and integrated circuits and gave these circuit and system designs at no cost to its customers. Unlike other silicon firms, Fairchild Semiconductor greatly emphasized the design of new applications in order "to broaden markets for existing products as well as to assist in developing markets for new products."²⁸ The application engineering department, again, played a major role in this effort. It concentrated particularly on writing application notes and designing prototypes of commercial equipment around the firm's devices.²⁹

Fairchild's application notes were meant to "show [commercial customers] how their problems could be solved better" with silicon components than with vacuum tubes, germanium transistors, and electromechanical components. The application notes described the functions commercial manufacturers could perform with Fairchild circuits and the systems they could make with the firm's components. They also explained in great detail how these circuits and systems could be designed. They included circuit diagrams, board layouts, and, in some cases, a mathematical analysis of the circuit or a short discussion of the system's mechanical design. A note on a "transistor stereo FM multiplex adapter," published in 1962, offers a good example: It showed how to build a new kind of adapter for multiplex and monophonic FM reception with Fairchild's diodes and transistors. Besides a comparative treatment of various demodulation techniques and their respective advantages and drawbacks, the note presented a diagram of the FM adapter and a detailed board layout in actual size. It also explained how to integrate the new device into stereo equipment.³⁰

Because of their creative designs and deft use of silicon components, the application notes sold the firm's devices. They also became

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instant engineering "classics" and were widely used by circuit and system engineers in the commercial sector. "People would design right out of the application notes," recalled Hulme, "they would take the application note and would essentially copy it."³¹ Used extensively by computer and consumer electronics firms in the United States, Fairchild's application notes were also copied by radio firms in Asia. Sony and most Hong Kong electronics firms closely modeled their transistor radios after Fairchild's notes. Indeed, Fairchild's application laboratory served de facto as the engineering department of many Hong Kong's radio manufacturers and introduced a stream of radio designs which these firms later produced and marketed.³²

In conjunction with the writing of application notes, engineers in the application laboratory built and demonstrated prototypes to convince commercial manufacturers of the system potential of Fairchild's devices. They designed and constructed a wide variety of electronics-based devices and systems such as burglar alarms, calculators, and radio transmitters. For example, they developed in the early 1960s a signalling system for the Ford Thunderbird to demonstrate that it could be built more easily and at lower cost with silicon devices than with electromechanical components. "We wanted that business," recalled a former marketing manager at Fairchild, "and we wanted to show that we could do that electronically. So we bought a replacement rear fender for a Thunderbird and we had this mounted in the conference room. These Ford executives came in and we made it do its thing, turn right, turn left based on electronics and, of course, it was a tiny package. That's how we tried to get people interested in our components."³³

Similarly, the application engineering laboratory designed the first all-transistor television set in 1962 to "demonstrate the feasibility of using transistors in TV." Along with notes on IF amplifiers and other television circuits, Fairchild's application engineers built a television prototype which was smaller, more reliable, and had better temperature characteristics than TV sets then available on the market. With the assistance of the firm's sales and marketing department, they later demonstrated it with the corresponding transistor kit to television manufacturers such as Zenith and General Electric. As the Fairchild design vastly "exceeded conventional television performance," it was widely adopted by the television industry and mass produced by General Electric, Zenith, and Sylvania. The consumer electronics industry's embrace of Fairchild's design created a very large and highly lucrative market for the firm and helped it

dominate the business for television transistors and integrated circuits well into the 1970s.³⁴

MANUFACTURING INNOVATIONS

In conjunction with the design of system-oriented products and the development of new applications, Fairchild Semiconductor scaled up production and drastically reduced prices in order to penetrate the commercial markets. Understanding that the adoption of silicon components in the consumer electronics and computer industries depended to a large degree on price cuts and increase in production volumes, Fairchild's management pursued a strategy of drastic price reductions. Not unlike Texas Instruments which a few years earlier had cut prices to create a commercial market for germanium devices, the firm announced a dramatic price cut on some of its planar transistors from \$5 to 25 cents in the Spring of 1963. One year later, Fairchild slashed prices on some of its integrated circuits from \$20 to \$2. In parallel to these radical price cuts, the firm continued to reduce prices on its product lines. Sometimes, it was selling components below manufacturing costs with the expectation that profitability would be obtained by reaching higher production volumes.³⁵

To drive costs down and increase production volume, Fairchild's management hired a group of engineers trained in mass production at Ford and General Electric. Between 1960 and 1966, these men gradually reshaped the organization and the techniques of production at Fairchild and transformed a small craft-oriented manufacturer into the first mass producer of planar diodes, transistors, and integrated circuits.³⁶ "My job as production manager," remembers Charles Sporck a former GE engineer and the architect of Fairchild's move to mass production,

was to get the increase in volume required but also to drive the cost down because the market that was available to us was very price dependent; and much of the very early business we got was military business which was not so price sensitive. The first orders we got from IBM were for \$150 a piece, for transistors. And literally some 7 or 8 years later we were selling those for less than 10 cents a piece, same devices. So there was an *enormous* amount of pressure to cut costs as well as get more volume.³⁷

To handle the growth in volume and cut manufacturing costs, Sporck and other mass production engineers introduced new organizational forms and production control techniques from Ford and General Electric. These men also scaled up production and drastically reduced manufacturing costs through yield improvement, throughput enlargement, partial automation and mechanization, and the establishment of off-shore assembly plants in the Far East.

Along with a new mindset characterized by cost consciousness and an emphasis on output, Sporck and his acolytes brought novel organizational structures and production control techniques. Sporck, who had gained his mass production expertise at General Electric's capacitor division, restructured the organization of production at Fairchild Semiconductor along the GE model. While manufacturing at Fairchild Semiconductor had previously been organized around process steps and was characterized by parallel organizations for production and process engineering, Sporck moved the company in a new direction by reorganizing the firm's production department along product lines. These product lines were built around families of products which used similar processes: small geometry NPN transistors, large geometry NPNs, PNPs, and, later, radio frequency transistors and integrated circuits.³⁸

Product lines were responsible for processing devices from wafer start to shipping as well as lowering manufacturing costs and improving the yields. Directed by a product engineer, they consisted of a production group and an engineering group. The production group was responsible for processing the devices to specifications and schedule. The engineering group maintained and troubleshot the process. These product lines also had their own diffusion and assembly areas, making them in effect mini-factories within the larger plant. While under the previous organization authority and responsibility had been diluted and relations between engineering and manufacturing were often antagonistic, the new structure centralized power and authority in the hands of the product engineer and helped "build [groups] which addressed problems as teams instead of adversaries."³⁹ Product lines made also for more efficient manufacturing as the equipment was used continuously and the engineering and production staff could gain experience in the manufacture of a few products.⁴⁰

The mass production engineers introduced also a new emphasis on production control. Sporck and other engineers from General

Electric centralized and strengthened the product planning function. More importantly, they introduced new production and inventory control techniques from General Electric to plan, schedule, and expedite the flow of parts and materials through the factory. In particular, they brought in new techniques for controlling wafer-in-process inventories. This was a growing problem as the number of wafers and processing steps increased dramatically in the early and mid-1960s. More generally, Fairchild Semiconductor's manufacturing engineers set up a very elaborate system to track the progression of wafer types, dice, and headers within the fab and the assembly lines. This was a task of substantial proportion as Fairchild processed more than 9,000 different components, some counting in the millions, at any given time in its factories by the mid-1960s.⁴¹

The new management techniques and organizational structures brought by Sporck and his associates made possible the move to mass production volumes as well as very substantial cost reductions in manufacturing at Fairchild Semiconductor. To achieve volume production at low cost, they established a program of continuous throughput enlargement and yield improvement in the wafer fab areas. "We solved little problems and scaled up," recalled a Fairchild engineer, "and [we] experimentally identified other problems, empirically attempted to solve them, sometimes contradicting what was then the acceptable knowledge."⁴² Emphasizing output, Sporck and the mass production-minded engineers continually increased the number of wafers in each batch. For instance, epitaxial reactors, used for deposition of semiconductor films over the silicon crystal, treated 16 wafers per load in 1962. They could process 26 wafers per load two years later. More importantly, while their predecessors had used 3/4 inch crystal ingots in the early 1960s, these men progressively enlarged the wafers' diameter to two inches, leading to a sevenfold increase in wafer surface areas. The increase in batch and wafer size made possible a very substantial increase in production volumes and had the added benefit of maximizing the use of capital equipment and therefore reducing unit costs.⁴³

In conjunction with continuous volume increases, Fairchild Semiconductor tightened its control of manufacturing processes as a way of improving yields. Yields were the largest cost determinant in the wafer fab area. Yields, or the percentage of fully operational units emerging from the production process, were very low in the early 1960s: they averaged 30% for planar transistors and were in the 5 to 10% range for integrated circuits. To improve yields, the product

line groups tightened their control of a very complex and unforgiving process. They achieved this through experimentation, trial and error, and by systematically applying statistical process control techniques to manufacturing problems. Although little is known about process control at Fairchild in the early and mid-1960s, it is clear that the firm's manufacturing engineers concentrated at first on photo-masking and solid state diffusion. These processes were the main determinants of yield loss in the early 1960s. Sources of defects in diffusion and photolithography such as misalignment in successive photoengraving steps were increasingly identified and remedied. As a result, yields increased rapidly. They plateaued in the mid-1960s however, leaving Fairchild's process engineering groups to look for other failure modes. Encountering substantial problems in producing large scale integration memory chips for Scientific Data Systems, these groups identified dust particles as a major yield determinant and controlled airborne contamination by establishing much tighter cleanliness standards in the fab area.⁴⁴

The firm also improved process control by systematically analyzing failure mechanisms in the research laboratory. Investigating the physics and chemistry underlying silicon processing, research groups at Fairchild made two important findings which had a major impact on yields. A group composed of Bruce Deal, Ed Snow, and Andy Grove identified positive mobile ions such as sodium as the cause of inversion layers in PNP and MOS transistors. These inversion layers were a major source of yield loss. Building on these findings, process engineers in the fab identified the ways in which mobile ions were introduced into the production process. They found out that the solvents routinely used to clean wafers and diffusion tubes contained sodium and thereby contaminated the wafers. By more carefully controlling the solvents' chemical composition, Fairchild's manufacturing engineers gradually eliminated the causes of sodium contamination and were able to produce PNP transistors and MOS integrated circuits with a stable silicon-silicon oxide interface.⁴⁵

Similarly, a critical problem in integrated circuit manufacturing in the early and mid-1960s was the formation of residues or "scums" after the removal of the photoresist. These residues prevented the proper etching of silicon oxide and led to very substantial yield losses. Focussing on this problem, a process development group made the surprising discovery that these residues appeared during smoggy days: ozone, they found out, would lead to the unwanted polymerization of the photoresist and the formation of residues. To prevent



DIODE FINAL SEAL

1959
240 Per Hour

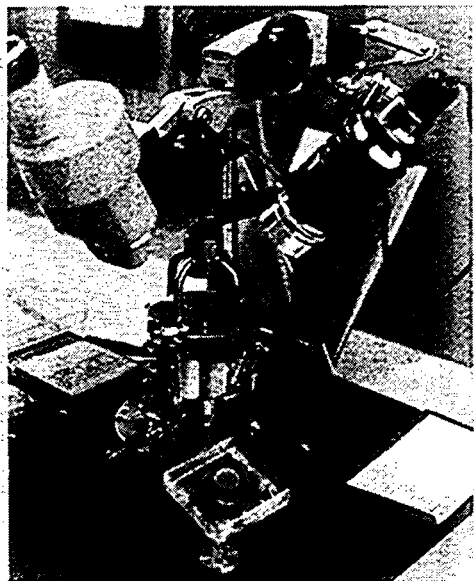
1964
2,000 Per Hour

Fig. 2 Diode sealing stations at Fairchild's San Rafael plant, 1959 and 1964.
Source: "Dr. Noyce Talk to FCI," December 1964, in box 1963-1964, Photographic Files of Steve Allen, Stanford Archives and Special Collections.

photoresist polymerization, Fairchild engineers dried resist-coated wafers in ozone free air or in nitrogen. More generally, to reduce exposure to smog, manufacturing engineers reduced work in process and speeded up the flow of batches throughout the fab. As a result of these and other findings as well as related problem solving in the fab, cumulative yields increased very significantly in the 1960s. They reached 60 to 80% for planar transistors and 50% for integrated circuits, making possible major cost reductions in the fab area.⁴⁶

In conjunction with their program of yield improvement and throughput enlargement in the wafer fab, Sporck and the mass production engineers devoted substantial attention to assembly and testing. This was the second part of the manufacturing process, where labor, rather than yield, was the largest cost determinant. To cut labor costs and enlarge production in the manufacture of diodes, Sporck and his associates resorted to automation. Diode final seal, a critical step in diode assembly, is a case in point. Diodes had been sealed one at a time in 1959, requiring considerable skill and manual work. By 1964, sealing was accomplished by rotary machines borrowed from the receiving tube industry. As diodes were automatically fed and sealed, the output of sealing stations grew 8 fold from 240 diodes per hour in 1959 to 2,000 diodes per hour in 1964.⁴⁷

In the case of transistors and integrated circuits, Fairchild Semiconductor resorted to scientific management techniques and off-shore manufacturing to reduce assembly costs. Sporck and other mass production-oriented engineers had first attempted to automate transistor assembly. An industrial engineering group in the transistor plant had developed a "semi-automatic assembly line" (SAAL) and tested it in production in 1960. These automation efforts were abandoned however in the next few years because the firm's semi-automatic lines still required substantial manual labor and did not "take nearly enough cost out of the assembly operation."⁴⁸ More importantly, Sporck and other Fairchild Semiconductor's managers realized that the firm did not have the engineering competence and the financial resources to fully automate transistor assembly, a task of unusual complexity. Unlike other silicon firms such as Texas Instruments and Motorola, Fairchild could not rely on much internal mechanical expertise. Neither could it find it readily in the area. Apart from precision machine shops brought about by the specialty tube industry, the San Francisco Peninsula's labor pool was lacking mechanical expertise. Furthermore, Fairchild Semiconductor's management felt that Fairchild Semiconductor, a firm with \$45 million



1961 Die Attach
90 Per Hour



1964 Die Attach
220 Per Hour

Fig. 3 Die attach stations at Fairchild's Mountain View plant in 1961 and 1964.
Source: "Dr. Noyce Talk to FCI," December 1964, in box 1963-1964, Photographic Files of Steve Allen, Stanford Archives and Special Collections.

in sales in 1963, did not have the financial resources necessary to develop very complex automatic lines.⁴⁹

Rather than "automating the labor out" in transistor and integrated circuit assembly, Fairchild's mass production-oriented engineers sought to increase the productivity and reduce the cost of manual labor for each device. In an effort to improve production efficiency, they maximized the output of individual stations in the assembly process by "[taking] small evolutionary steps." "We tried to determine what obstacles were in the way," recalled a Fairchild engineer, "and then on a miniature basis attack these specific things."⁵⁰ Die attach, the process by which the silicon die or chip is bonded to the package, is a good example. Fairchild's industrial engineers more than doubled the productivity of die attach stations between 1961 and 1964. While operators attached 90 dice per hour in 1961, the average output reached 220 in 1964. Fairchild's engineers achieved this increase in productivity by small and seemingly trifling adjustments in work stations. For instance, they brought metal headers closer to the die plate. They mounted a triangular plastic piece on the microscope to help the operator position herself better in relation to the microscope. And finally, they added an electronic mechanism which better controlled the temperature of the bonding material.⁵¹

As these improvements in productivity were not sufficient to reach "commercial" prices, Sporck delocalized Fairchild Semiconductor's assembly operations to low labor cost areas. His goal was to reduce overhead expenses as well as direct and indirect labor costs. In 1962, Sporck founded a plant in Maine where wages for unskilled operators were half of those in the Bay Area. One year later, in a move which was later widely copied by other silicon manufacturers, the firm established a subsidiary in Hong Kong to take advantage of the Territory's cheap labor. This decision met substantial opposition from the military. Fairchild Camera's board of directors also worried about Hong Kong's political instability and its proximity to Red China. Engineers doubted that such "a product could be produced in South East Asia."⁵² The labor cost differential however was so great that this opposition quickly quietened down. While the hourly rate for assemblers in the Bay Area was around \$2.80, its counterpart in Hong Kong was only 25 cents. Hong Kong not only offered low labor costs for unskilled operators but also cheap engineers eager to work in electronics. "There was a lot of engineering talent around," recalled Sporck, "and engineering talent that had nothing to do.

Many guys had gotten degrees in the United States and went back to Hong Kong and had nothing to do. So we had the opportunity of putting tremendously talented people to work on mundane things, but to them that was not so mundane compared to what they were working on normally."⁵³

Exploiting low labor costs to the fullest, Fairchild shifted in a few years most of its assembly and testing operations to Hong Kong. At first, the Hong Kong plant specialized in transistor assembly. It rapidly gained other functions such as die testing, final testing, and wafer sorting. It also increasingly assembled and tested integrated circuits. By 1966, Fairchild Semiconductor employed more than 4,000 operators, technicians, and engineers and was the Territory's largest electronics employer. As the Hong Kong plant rapidly ran out of capacity, Fairchild established another factory in South Korea in 1965 which specialized mostly in the assembly of integrated circuits. Because other silicon firms such as Texas Instruments and Motorola kept assembling their products in the United States, the Korean and Hong Kong plants gave Fairchild Semiconductor "a tremendous [competitive] advantage, one which was not really matched [by other firms] until ten years later."⁵⁴ More importantly, off-shore assembly helped Fairchild drastically reduce its manufacturing costs to meet those of radio tubes and germanium transistors.⁵⁵

PACKAGING INNOVATIONS

Off-shore assembly, rising yields, and enlarged throughputs made possible mass production volumes and a dramatic decrease in production costs. But as the Fairchild leadership found out, cost reductions in manufacturing did not suffice to achieve price parity with germanium devices, electron tubes, and electromechanical components. As labor costs went down through automation and off-shore assembly, packaging emerged as the largest part in the overall cost of silicon devices in the early and mid-1960s. In addition to being expensive to produce, packages traditionally used in military applications such as TO-5s and flatpacks (discussed below) were costly to attach to printed circuit boards. They also did not lend themselves to automatic insertion and soldering techniques which were widely adopted at that time in the computer and consumer electronics industry.

To get into the commercial market, "the principal problem we face," wrote Gordon Moore, the head of the R&D lab, in 1963, "is low cost packaging for microcircuits, transistors, and diodes."⁵⁶ To solve this problem, Fairchild Semiconductor developed two new innovative packages, plastic encapsulation and the dual in-line package (DIP). While the standard package for transistors had been a relatively costly metal can, Fairchild Semiconductor devised an inexpensive process to encapsulate transistors with epoxy. Under this process, dice were placed on a ceramic bead and covered with a plastic blob. While plastic encapsulation led at first to the contamination of the dice as well as hermeticity problems, the process had the advantage of relying on cheap materials and required very little capital expenditure. More importantly, because epoxy encapsulation required relatively little skill from the operators, it was easily performed in low-cost labor locations such as Hong Kong and South Korea. Along with off-shore assembly, the epoxy package permitted Fairchild to dramatically reduce its prices on planar silicon transistors and thereby match those of radio tubes and germanium transistors.⁵⁷

Engineers in the device development and digital system research sections of the R&D laboratory developed also the dual in-line package for integrated circuits. This was a major innovation which was widely adopted by the silicon industry in the mid-1960s and became the industry's package of choice well into the 1980s. The dual in-line package was meant to replace two military-oriented packages, the TO-5 and the flatpack. These packages were expensive and difficult to assemble. As a result, they hampered the adoption of integrated circuits by the commercial sector. The TO-5, a package made of a metal can and a metallic lid, had been widely used by Fairchild Semiconductor in its early integrated circuit families. This package however had its shortcomings. It had from eight to ten pins and did not allow the higher pin counts required by the firm's new diode transistor logic circuits. Moreover, the TO-5 was very expensive. Its round configuration made board layout difficult. It was also hard to attach to the printed circuit board as its leads had to be spread and bent out during system assembly.

Similarly, the flatpack had a number of limitations for commercial applications. The flatpack had been developed at Texas Instruments with support from the US Air Force to miniaturize electronics equipment. It was a small rectangular glass and ceramic package with close lead spacing and could be stacked easily. It also had good thermal dissipation characteristics. As a result, the flatpack became the package

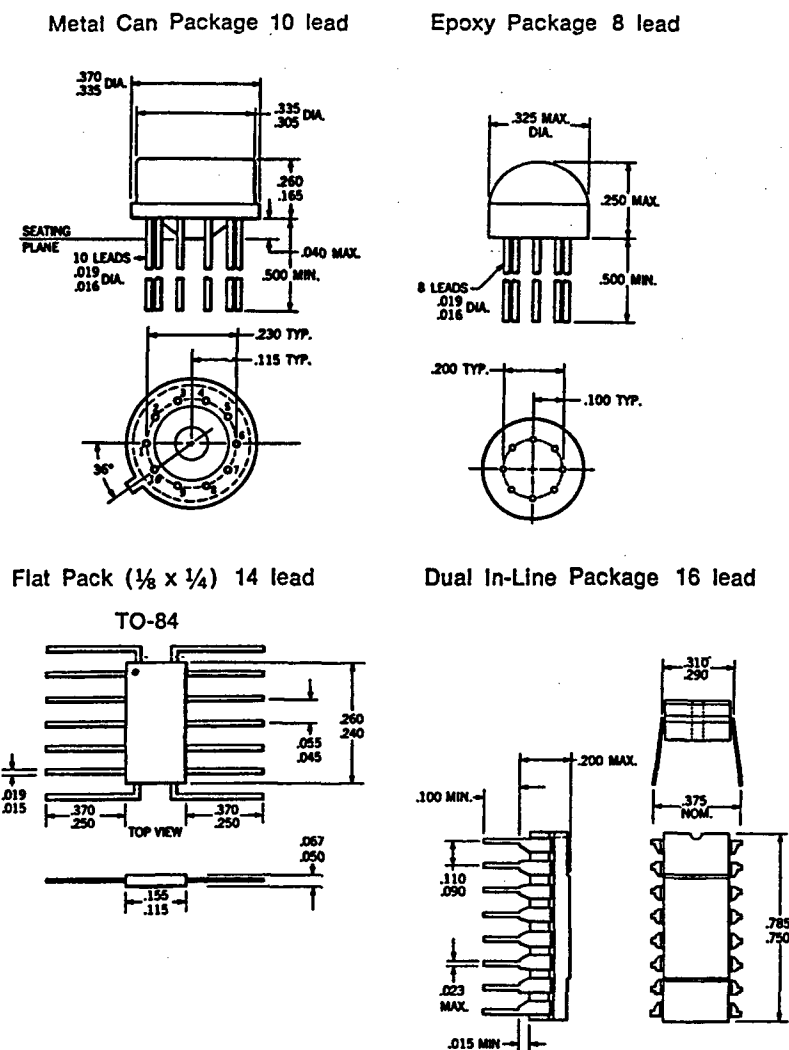


Fig. 4 Fairchild Semiconductor's TO-5, epoxy, flatpack, and dual in-line packages, 1966.

Source: E. Floyd Kvamme and L.H. Bieler, *Fairchild Semiconductor Integrated Circuits* (Mountain View, California: Fairchild Semiconductor, 1966), 80-81.

of choice for many military users and was later copied by other firms such as Fairchild. Flatpacks however, as Fairchild engineers discovered, did not meet the cost and manufacturing requirements of

users in the computer and consumer electronics industries. Flatpacks were expensive. More importantly, because their leads were placed very close to each other, 50 mils or thousandths of an inch apart, they required the use of printed circuit boards with narrow traces which were very difficult and expensive to produce. Furthermore, the use of flatpacks entailed very substantial labor costs. They had to be hand soldered and could not be automatically inserted into printed circuit boards. At this time, computer manufacturers were turning increasingly to automatic insertion machines to reduce their own manufacturing costs.⁵⁸

Because flatpacks and TO-5 packages were a major obstacle for the widespread use of integrated circuits in commercial applications, engineers in the digital systems and device development sections of the R&D department set out to develop a new package. They needed a package that would simplify board layout and could be soldered to cheap printed circuit boards. These men were also interested in designing a protective container which would lend itself to "easy industry use in automated insertion systems or very rapid hand insertion."⁵⁹ To facilitate printed circuit board layout and reduce the assembly costs of electronics systems, Rex Rice, the head of the digital systems laboratory, approached the development of new integrated circuit packages from a system perspective. He reasoned that design economy and ease of assembly depended on the packages' terminal configurations. As a result, Rice explored various lead arrangements. In particular, he devised a new lead configuration for integrated circuit packages whereby pins would be 100, rather than 50, mils apart. The pins were also in a line rather than coming from the package's four sides as in the flatpack or being in a circle as in the TO-5 package. To explore the effects of the in-line configuration on packing density and subsystem intraconnection, Rice laid out logic boards with different versions of the in-line arrangement and compared them with similar boards using flatpacks. "The trials," he reported,

were as follows: 1. Flat package with two rows of four opposed terminals. 2. Flat package with two rows of 4 terminals offset so interconnections could continue across the package between contacts on the opposite side. 3. In-line package with terminals arranged so no internal crossovers between die and terminal occurred. [...] 4. In-line package with terminals arranged so inputs and outputs alternated. [...] This sample of logic is certainly too small a trial to allow the drawing of sweeping conclusions.

However the following items were generally apparent: a. The flat packages took more logical layout design time than the inline packages. b. The flat packages took more two dimensional board area than the in-line packages. c. The 8 lead flatpack requires five horizontal runs on the average. The 8-lead in line package required an average of 2 extra horizontal runs.⁶⁰

Based on these findings, Rice, in collaboration with a packaging group in the device development section, designed a package that would not only simplify the layout of printed circuit boards but make their assembly more economic. After repeated trials and errors and consultations with a heat transfer specialist from Stanford, they settled on a single in-line design with 12 to 16 pins. Before transferring the in-line package to production, Fairchild's marketing department tested it on computer manufacturers and in particular on Control Data and General Electric. "One of the marketing guys," recalled Moore, "went around and tried it on customers. Customers did not like things standing up in one line. But they were willing to take two."⁶¹ As a result, Fairchild's device development group transformed the in-line prototype into a package with two parallel rows of pins.

The end result was a radically new package, one different from the flatpack. Whereas flatpacks were small and their leads spaced 50-mil apart, the dual in-line packages were larger and had wider lead spacing. This configuration facilitated the use of cheap printed circuit boards. Dual in-lines were easier to handle and more adaptable to automated assembly. They also plugged into the board and thereby could be soldered using mass production techniques. As they allowed for wire routing under the package, they led to a more efficient use of board space. Finally, their in-line arrangement simplified the layout of printed circuit boards. As a result, dual in-line packages were reported to "reduce assembly costs by a factor of four because of ease of use and automation capability."⁶² Introduced in 1965, the dual in-line package was embraced by computer manufacturers and was partly credited at the time for the explosion of integrated circuits in the commercial market in the late 1960s.⁶³

CONCLUSION

This essay has focused on Fairchild Semiconductor's expansion into the commercial markets and the changes in marketing, packaging,

and manufacturing that made it possible. It has argued that Fairchild entered new markets because of commercial exploitation motives and changes in military procurement policies. It also showed that the firm developed these markets by drastically reducing its prices and enlarging its production volumes. To do so, Fairchild integrated the mass production expertise from the electrical and automotive industries and delocalized its assembly operations to the Far East. Fairchild also hired circuit and system engineers from sectors originally intended to be users of its products. These engineers developed advanced components for commercial users. They also designed electronic systems around the firm's products which Fairchild gave at no cost to its customers. Finally, this essay has suggested that Fairchild's case was both representative and paradigmatic. Many silicon manufacturers such as Intel, National Semiconductor, and Texas Instruments, followed the firm into the computer and consumer electronics markets in the mid- and late 1960s. More importantly, Fairchild set a pattern for market entry into a wide range of industries in the following years. While Fairchild had mostly concentrated on the computer and consumer electronics industries, its spin-offs used Fairchild's sales and marketing techniques to build markets for a new silicon technology, MOS circuits, in a wide range of industrial sectors. In particular, they penetrated in the calculator, machine tool telecommunication, and automotive industries in the 1970s.⁶⁴

A number of lessons can be learned from Fairchild's, Intel's, and other firms' efforts to turn silicon electronics into a general purpose technology. First, the rise of the silicon industry to center stage of the post-war political economy can be partly explained by the innovative ways in which it interacted with user sectors. The silicon industry developed an unusual range of marketing competences. It also brought the customers in by hiring circuit and system engineers from a variety of commercial industries. Such tactics enabled silicon firms to acquire a strong system knowledge across a wide range of industries and to know nearly as much as its customers about circuit and system design. Partly because of its close contact with user sectors, the silicon industry showed considerable flexibility in developing and perfecting new skills, techniques, designs, and processes in response to its customers' needs. This expertise was translated into new products and the design of novel applications to attract new users. Fairchild, for instance, developed an all solid state television which was later manufactured with minor changes by Zenith and General Electric. Finally, silicon firms reshaped the interface between

electronics components and subsystems by developing new packages such as dual in-lines and plastic molded packages.

Second, manufacturing innovations played a crucial role in the rise of the silicon industry to become the center of US manufacturing. Hiring mass production engineers from the electrical, consumer electronics, and automotive industries, silicon firms were able to scale up their unique manufacturing systems and reach mass production volumes. Thereby, they invented a new form of mass production, different from the one in the mechanical industries. While silicon manufacturing shared some characteristics with mechanical modes of mass production such as low prices, single purpose machinery, and large volume production, it differed in a number of ways: silicon firms relied extensively on low cost labor in South-East Asia and used assembly lines in only part of their production processes. More importantly, semiconductor manufacturing was characterized by batch production in laboratory settings, tight industrial discipline, high engineering content, and considerable attention to precision and reliability.⁶⁵

The development of this new form of mass production at Fairchild Intel, National, and other silicon firms was akin in historical significance to the invention of mechanically-oriented forms of mass production at the Ford Motor Company in the early 1910s. Like Ford, Fairchild and other silicon corporations developed a new form of manufacturing which was later embraced by a wide range of industrial sectors. For example, the magnetics and disk drive industries adopted many mass production techniques and organizational forms from the silicon industry in the 1970s and early 1980s. More importantly, by manufacturing ever more complex integrated circuits in mass quantities, Fairchild and its followers laid the industrial infrastructure for the extraordinary growth of computing since the early 1960s, a phenomenon often referred to as the "information revolution."

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Notes

I mean by "silicon industry" the industry manufacturing silicon diodes, transistors, and integrated circuits.

1. Timothy Brasnihan and Manuel Trajtenberg, "General Purpose Technologies, 'Engines of Growth'?" *Journal of Econometrics*, 1995, 83–108; Nathan Rosenberg, "Technological Change in the Machine Tool Industry, 1840–1910," *Journal of Economic History*, 23 (1963), 414–443.
2. A.M. Golding, *The Semiconductor Industry in Britain and the United States: A Case Study in Innovation, Growth, and the Diffusion of Technology*, Ph.D. dissertation, Economics Department, University of Sussex, 1971; Richard Levin, "The Semiconductor Industry", in Richard Nelson, ed. *Government and Technical Progress: A Cross Industry Analysis* (New York: Pergamon Press, 1982), 9–100; William Edward Steinmueller, *Microeconomics and Microelectronics: Economic Studies of Integrated Circuit Technology*, Ph.D. dissertation, Economics Department, Stanford University, 1987.
3. For a good treatment of Shockley Semiconductor Laboratory, see Michael Riordan and Lillian Hoddeson, *Crystal Fire: The Birth of the Information Age* (New York: W.W. Norton, 1997), 225–253. For the establishment of Fairchild Semiconductor, see Gordon Moore, oral history interview conducted by Ross Bassett and Christophe Lécuyer, February 18, 1997; Jean Hoerni, oral history interview conducted by Lécuyer, February 4, 1996; Sheldon Roberts, oral history interview conducted by Lécuyer, July 6, 1996; Jay Last, oral history interview conducted by Lécuyer, April 1, 1996; Eugene Kleiner, oral history interview conducted by Lécuyer, May 21, 1996; Julius Blank, oral history interview conducted by Lécuyer, June 20, 1996; Victor Grinich, oral history interview conducted by Lécuyer, February 7, 1996.
4. Hoerni, "Semiconductor Device," U.S. patent 3,064,167, filed May 1, 1959, granted November 13, 1962; Hoerni, "Method of Manufacturing Semiconductor Devices," U.S. patent 3,025,589, filed May 1, 1959, granted March 20, 1962; Hoerni, "Planar Silicon Transistors and Diodes," Paper presented at the 1960 Electron Devices Meeting, Washington, D.C. – October 1960, Bruce Deal Papers, 88–033, Stanford Archives and Special Collections; Noyce, "Semiconductor Device-and-Lead Structure," US Patent 2,981,877, filed July 30, 1959; T.R. Reid, *The Chip: How Two American Companies Invented the Microchip and Launched a Revolution* (New York: Simon and Schuster, 1984), 76–78; Michael Wolff, "The Genesis of the Integrated Circuit," *IEEE Spectrum*, 13 (1976), 45–53. For an historical treatment of Fairchild's early history, see Christophe Lécuyer, "Making Silicon Valley: Engineering Culture, Innovation, and Industrial growth, 1930–1970." Ph.D. dissertation, History Department, Stanford University, 1999.
5. Kleiner, oral history interview conducted by Lécuyer, May 21, 1996, June 5, 1996, and June 16, 1998.
6. *Leadwire*, October 1959 and May 1960, courtesy of Jay Last.
7. One might surprise that Noyce's decision to enter the commercial markets was opposed by part of Fairchild Camera's management and possibly by other managers at Fairchild Semiconductor. Very little, however, is known about these conflicts. Nelson Stone, oral history interview conducted by Lécuyer, April 21, 1995; Pierre Lamond, oral history interview conducted by Lécuyer, November 17, 1995 and December 6, 1995; Hoerni, oral history interview conducted by Lécuyer, February 4, 1996; Roger Borovoy, oral history interview conducted by Lécuyer and Ross Bassett, June 14, 1996; Barry Miller, "Semiconductor Crisis Looms over Pricing," *Aviation Week*, June 12, 1961, 73–85; Barry Miller, "Competition to Tighten in Semiconductors," *Aviation Week & Space Technology*, September 23, 1963, 56–68; Lester Hogan, "Reflections on the Past and Thoughts about the Future of Semiconductor Technology," March 1977, volume 3, Bruce Deal Papers, 88–033, Stanford

- Archives and Special Collections; Don Hoeffler, "Silicon Valley USA," *Electronics News*, January 11, 1970, 1 and 4-5 and "Silicon Valley USA, Part II," *Electronics News*, January 18, 1970, 1 and 4-8; for an assessment of the competitive situation in planar transistors, see Progress Reports, Application Engineering, September 30, 1961, box 6, folder 8, Fairchild Semiconductor Papers, 88-095, Stanford Archives and Special Collections.
8. For a treatment of McNamara's reform of the procurement process, see Fred Kaplan, *The Wizards of Armageddon* (New York, 1983); special issue devoted to McNamara's policies, *Missiles and Rockets*, volume 12, March 25, 1963; Robert Freund, *Competition and Innovation in the Computer Industry*, Ph.D. Dissertation, Department of Economics, Duke University, 1971, 44.
 9. The military gave these drawings to other firms to obtain competitive bids. In the microwave tube industry, the DoD went one step further and forcefully created new suppliers by moving engineering teams from one company to the other.
 10. Philip J. Klass, "Defense to Speed Development, Cut Cost," *Aviation Week and Space Technology*, February 5, 1962, 26. For a treatment of the impact of the DoD's new procurement policies on component manufacturers, see Theodore Moreno, "Government Pricing and Procurement Policies," *The Microwave Journal*, January 1963, 165-166.
 11. A surge in manufacturing yields of the 2N706, a logic transistor, also contributed to Fairchild Semiconductor's severe inventory problems in 1960. Thomas Bay, "New Problems in Marketing Semiconductors," *The Solid State Journal*, August 1961, 55-56; Don Hoeffler, "I Didn't Raise My Boy to Be a Manager," *Electronics News*, October 17, 1966; Robert Freund, *Competition and Innovation in the Computer Industry*, Ph.D. Dissertation, Department of Economics, Duke University, 1971, 52; Victor Grinich, "Progress Report, Engineering Department," April 13, 1960, box 5, folder 2, Fairchild Semiconductor Papers, 88-095, Stanford Archives and Special Collections; Anonymous, "Transistor Entry Tightens," *Electronic News*, August 1, 1960, 1 and 4; Barry Miller, "Competition to Tighten in Semiconductors," *Aviation Week & Space Technology*, September 23, 1963, 56-68; Nelson Stone, oral history interview conducted by Lécuyer, April 21, 1995; Pierre Lamond, oral history interview conducted by Lécuyer, November 17, 1995 and December 6, 1995; and Grinich, oral history interview conducted by Lécuyer, February 7, 1996 and April 30, 1996; oral communication from Nelson Stone, June 2, 1998.
 12. Stone, oral history interview conducted by Lécuyer, April 21, 1995; oral communications from Stone, November 24, 1994 and June 2, 1998; Moore, oral history interview conducted by Ross Bassett and Lécuyer, February 18, 1997.
 13. Fairchild Semiconductor, *A Solid State of Progress*, ca. 1978, in 88-095, box 4, folder 9, Stanford Archives and Special Collections; Herbert Kleinman, interview with Robert Noyce, November 18, 1965, M827, Stanford Archives and Special Collections. Don Yost, oral history interview conducted by Lécuyer, August 12, 1997; Grinich, oral history interview conducted by Lécuyer, February 7, 1996; Oscar Reed, "Broadcasting Development Now Taking Place," *Proceedings of the IRE*, vol. 50 (May 1962), 837-847; A. Martinez, "Transistor Circuits in Television: Some Evolutionary Aspects," *The Radio and Electronic Engineer*, vol. 43, no. 1/2, January/February 1973, 103-114.
 14. Marshall Cox, oral history interview conducted by Christophe Lécuyer, October 25, 1996.
 15. R.W. Allard, "Product Line History Control Data EDP Systems," July 21, 1980, Charles Babbage Institute; James E. Thornton, *Design of a Computer: The Control Data 6600* (Scott, Foresman and Company, 1970); James E. Thornton, "The CDC 6600 Project," *Annals of the History of Computing*, vol. 2, no. 4 October 1980, 338-348; Kenneth Flamm, *Creating the Computer: Government, Industry, and High Technology*

- (Washington, D.C: Brookings Institution, 1988); Cox, oral history interview conducted by Lécuyer, October 25, 1996; Floyd Kvamme, oral history interview conducted by Lécuyer, September 25, 1996; Pierre Lamond, oral history interview conducted by Lécuyer, November 17, 1995, December 6, 1995, and August 12, 1997. For a treatment of computer packaging in the 1960s, see Charles Harper (ed.), *Handbook of Electronic Packaging* (New York: McGraw-Hill, 1969) and Jack Staller, "The Packaging Revolution, Part I: Form and Function Interact," *Electronics*, vol. 38, October 18, 1965, 72-87 and "The Packaging Revolution, Part II: Design and Manufacturing Overlap," *Electronics*, vol. 38, November 1, 1965, 75-87.
16. More than 6,000 of these packages were used in the CPU and memory of the 6600.
 17. Thomas Bay, oral history interview conducted by Christophe Lécuyer, July 2, 1996.
 18. Bay, oral history interview conducted by Lécuyer, July 2, 1996; Pierre Lamond, oral history interview conducted by Christophe Lécuyer, November 7, 1995, December 6, 1995, and August 12, 1997.
 19. Kenneth Flamm, *Creating the Computer: Government, Industry, and High Technology* (Washington, D.C: Brookings Institution, 1988).
 20. Moore and Grinich, "R&D Progress Report," December 12, 1961, box 6, folder 9, in 88-095, Stanford Archives and Special Collections; Moore, oral history interview conducted by Bassett and Lécuyer, February 18, 1997; Grinich, oral history interview conducted by Lécuyer, February 7 and April 23, 1996; Jack Gifford, oral history interview conducted by Lécuyer, July 23, 1996; Tom Bay, oral history interview conducted by Lécuyer, July 2, 1996.
 21. Bay, oral history interview, July 2, 1996.
 22. Gifford, oral history interview, July 23, 1996; Cox, oral history interview, October 25, 1996; Kvamme, oral history interview, September 25 and October 10, 1996. Fairchild Semiconductor also established a network of distributor representatives to pursue the smaller, often commercial, customers in 1961. For the establishment of distributor representatives by Fairchild Semiconductor, see Bay, oral history interview, July 2, 1996 and Bay, "New Problems in Marketing Semiconductors," *The Solid State Journal*, August 1961, 55-56.
 23. Hulme, oral history interview, March 18, 1997 and May 7, 1997; Moore, oral history interview, February 18, 1997. "Application Engineering Personnel," 1964, courtesy of John Hulme. Because of its strong system expertise, Fairchild knew nearly as much as the customers about their own systems. As a result, there was considerable anxiety among users about Fairchild entering the system business.
 24. Hulme, oral history interview, March 18, 1997.
 25. As a result, Fairchild was frequently sole source for critical products until competitors caught up. For example, automatic gain control (AGC), a feature critical to the design of intermediate frequency (IF) for radio and television applications, was relatively easy to obtain, using pentode tubes. Silicon transistors did not readily perform this function until Fairchild developed a special AGC transistor which made all transistor IF strips practical. It was several years before Fairchild's competitors were able to design and manufacture transistors which would allow gain control. Oral communications from Hulme, January 13, 1997 and April 8, 1999; Hulme, oral history interview, March 18, 1997 and May 7, 1997.
 26. "Dr. Noyce Presentation to FCI," December 1964, in box 1963-1964, Photographic Files of Steve Allen, 91-167, Stanford Archives and Special Collections; Floyd Kvamme and L.H. Bieler, *Fairchild Semiconductor Integrated Circuits* (Fairchild Semiconductor, 1966); Bay, oral history interview, July 2, 1996; Hulme, oral history interview, March 18 and May 7, 1997. For work on the automatic gain control transistors, see Moore and Grinich, "R&D Progress Report," April 13,

- 1962, box 7, folder 3; Moore and Grinich, "R&D Progress Report," May 11, 1962, box 7, folder 4; Moore, "R&D Progress Report," July 6, 1962, box 7, folder 6; Moore and Grinich, "R&D Progress Report," April 13, 1962, box 7, folder 4; Moore, "R&D Progress Report," August 9, 1962, box 7, folder 5; Moore, "Research and Development Progress Report for 1962," January 24, 1963, box 7, folder 6; Progress report, Device development section, July 1, 1962, box 7, folder 5, Fairchild Semiconductor Papers.
27. Fairchild, emphasized the development of concepts which could go in standard circuits as in the speed ECL, CTL, DCTL families. Hulme, oral history interview, March 18 and May 7, 1997; oral communication from Hulme, April 8, 1999; data sheets for the 2N709 and the 2N2368, Fairchild folder, Computer ephemera collection, CBI 12, Charles Babbage Institute. For Fairchild's difficulties in producing the CDC transistor, see Lamond, oral history interview, November 17, 1995, December 6, 1995, and August 12, 1997; Roger Smullen, oral history interview conducted by Lécuyer, May 12, 1996.
 28. "Fairchild Semiconductor," 1967, Ephemera collection, Graduate School of Business, Stanford University.
 29. Kvamme, oral history interview, September 25 and October 10, 1996.
 30. Larry Blaser, "A Transistor Stereo FM Multiplex Adapter," APP-44, March 1962, in Fairchild Semiconductor, *Product Catalog*, 1962, courtesy of Jay Last; Kvamme, oral history interview, September 25 and October 10, 1996; Hulme, oral history interview, March 18 and May 7, 1997. Fairchild's application engineering laboratory concentrated on radio, television, and computer applications. It published no less than 16 application notes in television engineering between 1961 and 1970. Besides producing detailed application notes, Fairchild Semiconductor also published on a newsletter, *Fan Out*, which introduced the firm's new devices and offered short descriptions of possible applications.
 31. Hulme, oral history interview, May 7, 1997.
 32. Hulme, oral history interview, March 18 and May 7, 1997; communication from Hulme, January 13, 1997; Kvamme, oral history interview, September 25, 1996; Gifford, oral history interview, July 23, 1996; John MacDougall, "Low Cost Semiconductors for the Consumer Market," *Electronics World*, 78 (3), September 1967, 37-39.
 33. Kvamme, oral history interview, October 10, 1997.
 34. Progress Reports, Application Engineering, August 1, 1961, box 6, folder 6; August 31, 1961, box 6, folder 7; September 30, 1961, box 6, folder 8; October 31, 1961, box 6, folder 8 - all in Fairchild Semiconductor Papers, 88-095, Stanford Archives and Special collections; Richard Lane, "A Synchronously Tuned 60 MC I.F. Amplifier with A.G.C.," APP-26, circa 1961, in Fairchild Semiconductor, *Product Catalog*, 1962, courtesy of Jay Last; Anonymous, "All Transistor TV," *Leadwire*, Fall 1962, courtesy of Michael Brozda, National Semiconductor; Cox, oral history interview, October 25, 1996; Hulme, oral history interview, March 18, 1997; Kvamme, oral history interview, October 10, 1997; Yost, oral history interview, October 12, 1997; Bay, oral history interview, July 2, 1996.
 35. Robert Freund, *Competition and Innovation in the Transistor Industry*, Ph.D. dissertation, Economics Department, Duke University, 1971, 62-64; "Fairchild Bids for Commercial Integrated Circuit Market," *Palo Alto Times*, May 4, 1964, collection of the author. For later price cuts, see "Manufacturers Outlook," *EDN*, volume 11, July 1966, 56-59; Cox, oral history interview, October 25, 1996.
 36. There is evidence that this process was not without conflicts. Engineers who had built up the manufacturing organization at Fairchild Semiconductor in the late 1950s seem to have resisted Sporck's approach to transistor manufacturing and subsequently left the firm. Eugene Kleiner, one of the firm's founders and head of manufacturing engineering, is a case in point. In an interview he recalled

conflicts with Sporck regarding the proper approach to transistor manufacturing: "We hired some manufacturing foremen, some manufacturing people, I think a fellow named Grady and then Charley Sporck and in my opinion they were good manufacturing people but they had no feel for transistor manufacture. And I felt that since I grew up with the whole product, I had a better feel for making good transistors than Charley Sporck. For instance he came from capacitor manufacturing at General Electric where they made, I don't know, how many millions of capacitors and you know we were not making capacitors here. We were making something much more delicate, much more difficult than a capacitor. There was a little friction in that area." Kleiner left Fairchild Semiconductor in 1961 to found Edex, a manufacturer of teaching machines. He later established Kleiner Perkins, one of the largest and most influential venture capital firms in the Silicon Valley. Eugene Kleiner, oral history interview conducted by Christophe Lécuyer, May 21, 1996, June 5, 1996, and June 16, 1998.

37. Charley Sporck, oral history interview conducted by Lécuyer, May 15, 1995.
38. Fairchild Semiconductor might have been the first firm in the silicon industry to adopt product line forms of organization.
39. Sporck, oral history interview, May 15, 1995.
40. Lamond, Oral history interview, November 15, 1995, December 6, 1995, and August 12, 1997.
41. Sporck hired one of his subordinates at General Electric's power capacitor operations, Jack Magarian, to install a new production control system at Fairchild Semiconductor in 1961. Sporck, oral history interview, May 15, 1995; Smullen, oral history interview, May 12, 1996.
42. Yost, oral history interview, August 12, 1997.
43. Kleiner, oral history interview, May 21, 1996 June 5, 1996, and June 16, 1998; Sporck, oral history interview, May 15, 1995; Smullen, oral history interview, May 12, 1996; "Dr. Noyce talk to FCI," box: 1963-1964, 91-167, Stanford Archives and Special Collections. The effort to increase wafer sizes led to innovations in the growing of silicon crystals; see E.M. Reithard, "Growing Two-Inch Silicon Crystals," *SCP and Solid State Technology*, May 1967, 46-47.
44. Yost, oral history interview, August 12 and September 23, 1997; Smullen, oral history interview, May 12, 1996. For a treatment of process control at Fairchild Semiconductor in the mid-1960s, see Fairchild Semiconductor, "Reliability '65," Fairchild Semiconductor folder, Computer ephemera collection, CBI 12, Charles Babbage Institute.
45. For the research laboratory's renewed interest in manufacturing problems from 1962 on, see Moore, "Monthly Progress Report," August 9, 1962, box 7, folder 5, Fairchild Semiconductor Papers. For research on the silicon-silicon oxide interface and efforts to eliminate mobile ions from the production process, see A. Grove, O. Leistiko, and C. Sah, "Redistribution of Acceptor and Donor Impurities During Thermal Oxidation of Silicon," *Journal of Applied Physics*, vol. 35, No. 9, September 1964, 2695-2701; B. Deal, A. Grove, E. Snow, and C. Sah, "Observation of Impurity Redistribution During Thermal Oxidation of Silicon Using the MOS Structure," *Journal of the Electrochemical Society*, Vol. 112, No. 3, March 1965, 308-314; B. Deal, A. Grove, E. Snow, and C. Sah, "Recent Advances in the Understanding of the Metal-Oxide-Silicon System," *Transactions of the Metallurgical Society of AIME*, volume 233, March 1965, 524-533 B. Deal, E. Snow, and A. Grove, "Properties of the Silicon DioxideSilicon System," *SCP and Solid State Technology*, vol 8, number 11, 1966, 25-33; B. Deal, M. Sklar, A. Grove, and E. Snow, "Characteristics of the Surface-State Charge of Thermally Oxidized Silicon," *Journal of the Electrochemical Society*, vol. 114, No. 3, March 1967, 266-274; Smullen, oral history interview, May 12, 1996.

46. The same process development group made another important innovation, chrome masks, which became one of Fairchild's most closely guarded trade secrets in the early and mid-1960s. Instead of covering the mask with a soft emulsion which would rapidly deteriorate in contact with the silicon wafer, they deposited and etched chrome on the glass plate. This innovation helped improve yields very substantially. For research on photoresist "scums," see "Progress Reports, Microelectronics Research Section," March 1, 1962, box 7, folder 2; August 1, 1963, box 8, folder: May–August 1963; September 1, 1963, box 8, folder: May–August 1963; October 1, 1963, box 8, folder 3; January 1, 1964, box 9, folder: September–December 1963; March 1, 1964, box 9, folder 2; April 1, 1964, box 9, folder 3; June 1, 1964, box 10, folder 2; "Progress Reports, Materials and Processes Department," September 1, 1964, box 10, folder 5; October 6, 1964, folder 7, box 10 – all in Fairchild Semiconductor Papers, Stanford Archives and Special Collections. For the work on chrome masks, see Moore, "R&D Monthly Progress Report," July 6, 1962, box 7, folder 6 and "Research & Development Progress Report for 1962," January 24, 1963, box 7, folder 6, all in Fairchild Semiconductor Papers; Grinich, oral history interview, April 23, 1997.
47. The automatic final sealer was developed by an engineer who had formerly worked at CBS-Hytron, a manufacturer of receiving tubes; Anonymous, "Final Sealer Will Speed Production," *Leadwire*, July 1960, 2, courtesy of Jay Last; Lamond, oral history interview, August 12, 1997; Yost, oral history interview, September 23, 1997; communication from Fred Bialek, August 26, 1997.
48. Smullen, oral history interview, May 12, 1996.
49. Anonymous, "Major Price Drop Announced; Mechanization Will Cut Costs, Lower Prices," *Leadwire*, March 1960, 1 and 4 and "Tooling Designs Semiautomatic Stations; Mechanized Line to Operate by Fall," *Leadwire*, July 1960, 1 and 5, courtesy of Jay Last; Smullen, oral history interview, May 12, 1996; Yost, oral history interview, August 12 and September 23, 1997; Sporck, oral history interview, May 15, 1995.
50. Yost, oral history interview, August 12, 1997.
51. Yost, oral history interview, August 12 and September 23, 1997.
52. Sporck, oral history interview, May 15, 1995.
53. Fairchild Semiconductor was not the first electronics firm to establish a manufacturing plant in Hong Kong. Other firms such as Ampex had core stringing operations there. But Fairchild was the first silicon corporation to do so. According to most accounts, the idea of setting up an assembly plant in Hong Kong came from Robert Noyce, one of Fairchild Semiconductor's founders and its general manager. Noyce had invested in a small radio firm in Hong Kong which alerted him to the low cost of direct labor and the availability of cheap engineering manpower there. Besides its interest in the Territory's low cost labor, Fairchild also established the Hong Kong assembly plant because of labor shortages in the Bay Area. As the area was growing very rapidly from a manufacturing point of view, it became more and more difficult to find operators. Furthermore, the competition for hourly workers with silicon experience grew rapidly as new spin-offs from Fairchild Semiconductor such as Amelco, Signetics, and GME started to hire operators from the firm. One might also speculate that the move to the Far East was prompted by unionization efforts in the Bay Area in the early 1960s and by the fact that Asian labor was deemed easier to lay-off than domestic workers. Finally, more and more consumer electronics firms, Fairchild's customers, were establishing assembly plants in the Far East. For instance, General Instruments built a plant to assemble television tuners in Taiwan in 1963. Sporck, oral history interview, May 15, 1995; Stone, oral history interview, April 21, 1995; Julius Blank, oral history interview conducted by Lécuyer, June 20, 1996; communication from Fred Bialek, August 26, 1997.

54. Stone, oral history interview, April 21, 1995.
55. For a treatment of electronics component manufacturing in Hong Kong in the 1960s, see Edward K.Y. Chen, *The Electronics Industry of Hong Kong: An Analysis of its Growth*, M. Soc. Sc. Thesis, University of Hong Kong, 1971.
56. Moore, "Research and Development Progress Report for 1962," January 24, 1963, in box 7, folder 6, Fairchild Semiconductor Papers, Stanford Archives and Special Collections.
57. Plastic packages became the package of choice for consumer applications and later, as Fairchild Semiconductor gathered substantial reliability data, for computer applications as well. While Fairchild used at first epoxy encapsulation for transistors, it also introduced an epoxy package for its RTL integrated circuits in 1964. Bay, oral history interview, July 2, 1996; communication from Bay, October 2, 1997; Yost, oral history interview, August 12 and September 23, 1997; John MacDougall, "Low Cost Semiconductors for the Consumer Market," *Electronics World*, 78 (3), September 1967, 37-39. For early work at Fairchild on plastic packages, see Moore to Noyce, January 8, 1960, box 5, folder 3; Moore and Grinich, "R&D Progress Report," August 11, 1961, box 6, folder 7; "Progress Reports, Chemistry Section," January 1, 1960, February 1, 1960, March 1, 1960, and April 1, 1960 - all in box 5, folder 1, Fairchild Semiconductor Papers, Stanford Archives and Special Collections. For the use of epoxy package for integrated circuits, see Cox, oral history interview, October 25, 1996 and Kvamme and Bieler, *Fairchild Semiconductor Integrated Circuits*, 80.
58. Moore, "Research and Development Progress Report for 1962," January 24, 1963, box 7, folder 6, Fairchild Semiconductor Papers, Stanford Archives and Special Collections; R.E. Koeper, "Microcircuit Packaging Evolution," *EDN*, vol. 11, September 28, 1966, 52-65. Automatic insertion machines and wave soldering were developed in the 1950s and introduced in the commercial computer industry in the early 1960s, see Kenneth Flamm, *Creating the Computer: Government, Industry, and High Technology* (Washington, D.C.: Brookings Institution, 1988); R.R. Batcher, "Product Engineering and Production," *Proceedings of the IRE*, vol. 50 (May 1962), 1289-1305; and S.F. Danko, "Printed Circuits and Microelectronics," *Proceedings of the IRE*, vol. 50 (May 1962), 937-945.
59. Rex Rice, "Progress Report, Digital Systems Research Department," November 3, 1963, box 11, folder 1, Fairchild Semiconductor Papers.
60. Rice, "Progress Report, Digital Systems Research Department," December 3, 1963, box 9, brown folder, Fairchild Semiconductor Papers.
61. Moore, oral history interview, February 18, 1997.
62. Bryant Rogers, "IC Packages: Which One To Use?" *Solid State Technology*, September 1968, 45-46.
63. A sandwich structure, the dual in-line package developed in R&D, was made of an etched lead-preform embedded in a "hard" glass between two thin ceramic sheets. The top sheet had a hole punched through it to expose the leads for bonding and assembly. It was covered with a ceramic cap during final seal. The package was later thoroughly redesigned by two production engineers Jack Gates and Roger Smullen to make it manufacturable. Fairchild Semiconductor later introduced a plastic dual in-line which further reduced packaging costs. For early work in R&D on the in-line package, see "Progress Reports, Digital Systems Research Department," October 1, 1963, box 9, brown folder; November 1, 1963, box 11, brown folder; December 3, 1963, box 9, brown folder; January 6, 1964, box 9, brown folder; March 3, 1964, box 9, folder 2; April 1, 1964, box 9, folder 3; May 5, 1964, box 10, folder 1; June 3, 1964, box 10, folder 2; June 30, box 10, folder 3; November 3, 1964, box 11, folder 1; August 7, 1964, box 10, folder 4; September 2, 1964, box 10, folder 6; October 1, 1964, box 10, folder 6 - all in Fairchild Semiconductor Papers. See also "Progress Reports, Device Development Section,"

- July 1, 1964, box 10, folder 3; August 1, 1964, box 10, folder 4; September 1, 1964, box 10, folder 5 in Fairchild Semiconductor Papers. For the redesign of the package from a single to a dual in-line configuration, see "Progress Reports, Digital Systems Research Department," November 3; 1964, box 11, folder 1; December 2, 1964, box 11, folder 1; January 4, 1965, box 11, folder 2; "Progress Reports, Device Development Section," November 1, 1964, box 10, folder 7; December 1, 1964, box 11, folder 1; January 1, 1965, box 11, folder 2—all in Fairchild Semiconductor Papers. For a treatment of the adoption of the dual in-line, see Kvamme and Bieler, *Fairchild Semiconductor Integrated Circuits*; Anonymous, "Microelectronics Goes Commercial," *EDN*, vol. 11, July 1966.
64. For a treatment of the history of MOS technology, see Ross Bassett, "New Technology, New People, New Organizations: The Rise of the MOS Transistor, 1945–1975," Ph.D. Dissertation, Princeton University, 1998.
65. For a treatment of the invention of mass production at the Ford Motor Company, see David Hounshell, *From the American System to Mass Production, 1800–1932* (Baltimore: Johns Hopkins University Press, 1984).